NSO8062 Geofüüsikaline andmetöötlus

Loeng 5: 1D interpoleerimine, korrelatsioon

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Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	Kuumalained	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

3. loengu tagasiside

- Halokliini leidmine: Kuna tegu on andmetega, mis on mõõdetud 0.5 dbar vertikaalse intervalliga (~iga 0.5 meetri järel), siis:

  1. Silume soolse andmed üle 2.5 m järgmise valemiga:

  SA_movav(i) = mean(SA_gkg(i-2:i+2)); % Kohale i keskmistatakse soolsuse väärtusi, mis asuvad kaks kohta ‘tagasi’ (i-2) kuni (:) kaks kohta ‘edasi’ (i+2).

  2. Kasutame iga sügavuse jaoks silutud sooluse andmete põhjal gradiendi väärtuse leidmiseks järgmist valemit:

  SA_grad_gkgm(i) = (SA_movav(i+2) – SA_movav(i-2)) / (pressure_dbar(i+2) – pressure_dbar(i-2));

  Vt. näidet Google Drive’is (finding_halocline_example.xlsx) kui vaja.

  Lae alla Google Drive’ist **Keri_2020_clean.mat** kui vaja.

  Lae alla Google Drive’ist **results_lec4_ex2.mat** kui vaja.

Interpolation – interp1 function

what we know		what we want to know	
pressure_dbar	SA_gkg	pressure_dbar	SA_gkg
x	**v**	**xq**	**vq**
1	4.7		
		2	interp1(x,v,2)
3	5.1		
4	5.2		
		5	interp1(x,v,5)
6	5.5		

vq = interp1(x,v,xq)

Returns interpolated values (vq) of a 1-D function at specific query points.

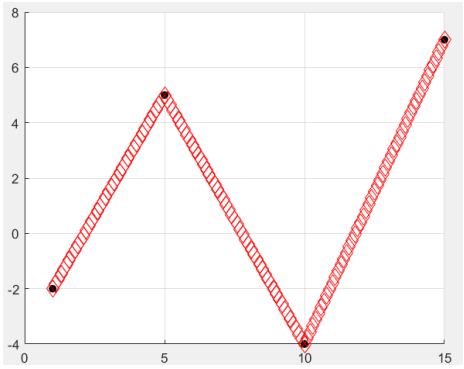
Vector x contains the sample points (e.g., measurement depths)

Vector v contains the corresponding values, v(x). (e.g., parameter values at measured depths)

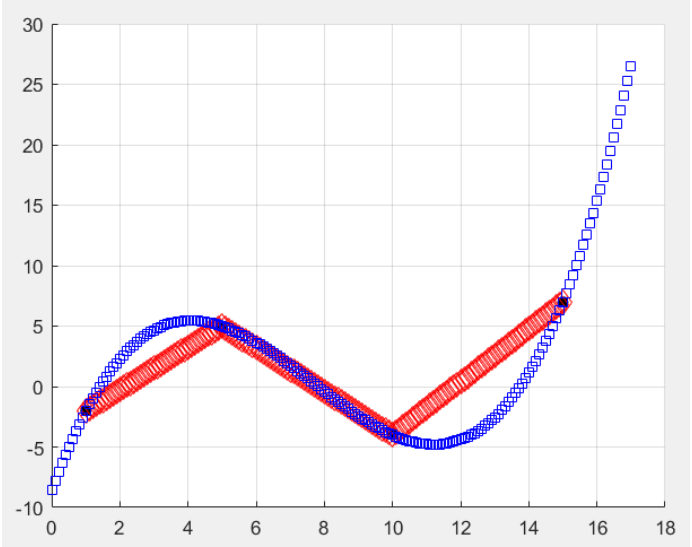
Vector xq contains the coordinates of the query points. (e.g., the depths for which we need the parameter values for)

By default, interp1 uses linear interpolation.

fillmissing – good to use on data, where we have NaNs
[https://se.mathworks.com/help/matlab/ref/fillmissing.html]



Interpolation – interp1 function



vq = interp1(x,v,xq,’spline’)

Returns interpolated values (vq) of a 1-D function at specific query points.

Vector x contains the sample points

Vector v contains the corresponding values, v(x).

Vector xq contains the coordinates of the query points.

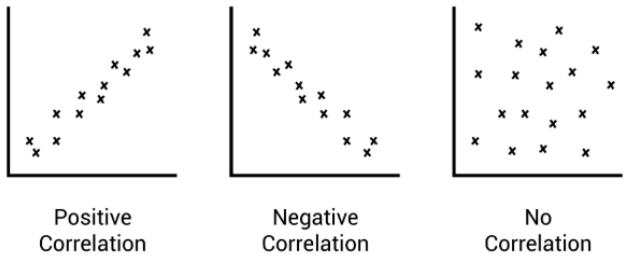
If **Linear interpolation** is like connecting the dots with a stiff wooden ruler (creating sharp, ugly corners), **Spline interpolation** is like using a thin, flexible piece of plastic that you bend to touch every point.

Imagine you have three points on a graph. To connect them, MATLAB doesn't just draw straight lines. Instead:

- It looks at the slope of the data coming into a point.
- It looks at the slope leaving that point.
- It uses a Cubic Polynomial (looks like a stretched-out "S") to make sure the transition is perfectly smooth, like a rollercoaster track.

[https://se.mathworks.com/help/matlab/ref/double.interp1.html]

Correlations between data



Bridge between simply collecting numbers and actually understanding data and explaining processes etc.

In a statistical sense, you are likely looking for the Pearson Correlation Coefficient (r). It ranges from -1 to +1:

- r = 1: Perfect positive correlation (as one goes up, the other goes up).
- r = -1: Perfect negative correlation (as one goes up, the other goes down).
- r = 0: No linear relationship at all.

Feature	Pearson Correlation (r)	Coefficient of Determination (R^2)
Range	-1 to +1	0 to 1 (or 0% to 100%)
Shows Direction?	Yes. (Positive or negative slope)	No. (Squaring a negative makes it positive)
Primary Goal	Measures the strength and direction of a linear relationship.	Measures how much variance is explained by the model.
Intuition	"How closely do these points hug a line?"	"How much of the 'noise' in Y is explained by X?"

r is for exploration (finding out if a relationship exists).

R^2 is for prediction (finding out how reliable your "rule" or model actually is).

Lecture 5. Exercise 1.

Keri_2020_clean.mat

- Select one profile. Remove data from the halocline layer.
- Interpolate missing data using interp1 and interp1 ‘spline’.
- Visualize all three profiles (original and two interpolated).

Add descriptions to your code and save code.

vq = interp1(x,v,xq)

vq = interp1(x,v,xq,’spline’)

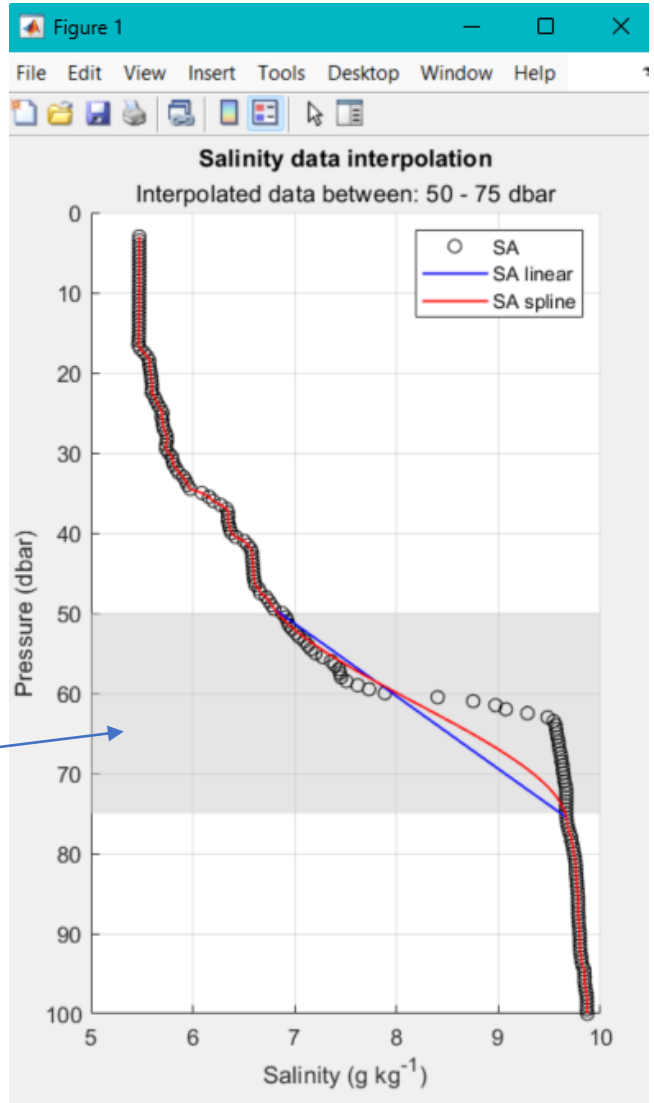
scatter, plot

set(gca,'YDir','reverse’)

xlim, patch, EdgeColor, FaceAlpha

xlabel, ylabel, title, subtitle, legend

gcf, Position



Lecture 5. Exercise 2.

results_lec4_ex2.mat

Table containing data on halocline layer maximum depth (the lower border) and oxygen values measured at deepest depth for each profile.

Get basic statistics of both variables (mean, median, min, max, std).

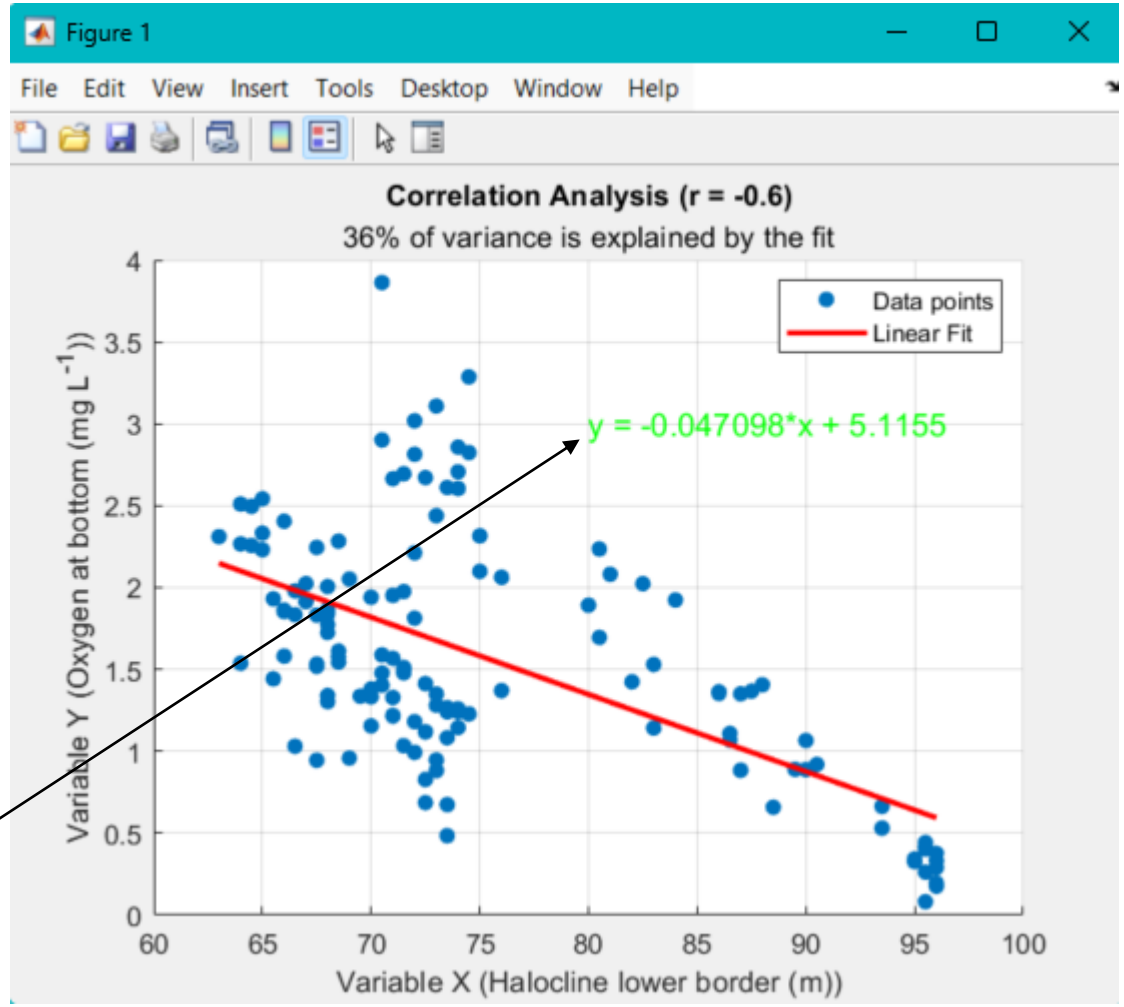
Get r, R^2, and p-value.

```
[R_matrix, P_matrix] = corrcoef(x,y)
r = R_matrix(1,2);
p_value = P_matrix(1,2);
```

Linear fit for visualization (polyfit, polyval)

```
p = polyfit(x, y, 1); % p(1) is slope, p(2) is intercept
y_fit = polyval(p, x); % Linear Fit
```

Add text to figure: y = p1*x+p2, where p1 and p2 are replaced by slope and intercept values from polyfit.



Add descriptions to your code and save code.

Lecture 5. Exercise 3.

$$T = \beta_0 + \beta_1 S + \beta_2 S^2$$

Keri_2020_clean.mat

Scatter salinity(x) and temperature (y).

Fit a 2nd (3rd) degree polynomial to see if there is a curve in data.

```
stats = fitlm(x,y,'poly2');
R_squared = stats.Rsquared.Ordinary;
coef = stats.Coefficients;
```

	1 Estimate	2 SE	3 tStat	4 pValue
1 (Intercept)	146.9089	0.4666	314.8513	0
2 x1	-48.3854	0.1889	-256.1840	0
3 x1^2	5.3797	0.0248	217.1792	0

(Intercept): This is β_0. It is the theoretical temperature if salinity were zero. (In oceanography, this is often just a mathematical anchor and doesn't always have a physical meaning).

x1: This is β_1. it represents the linear relationship. If this is negative, temperature generally drops as salinity increases.

x1^2: This is β_2. This is the curvature. If this value is significant, it proves the relationship isn't a straight line.

Estimate: The actual weight (coefficient) assigned to that variable. These are the numbers you'd use to write the equation out by hand.

SE (Std Error): How much "wiggle room" or uncertainty is in that estimate. Smaller is better.

tStat: The Estimate divided by the SE. It measures how many standard deviations the estimate is away from zero.

pValue: The most important column. It tells you if that specific part of the model is actually "real" or just noise (looking for pValue < 0.05).

Lecture 5. Exercise 3. (continued)

We have TS scatter plot, and the model (stats) for our data.

Let's create a salinity range and predict temperature.

```
S_range = linspace(min(S),max(S),100);
```

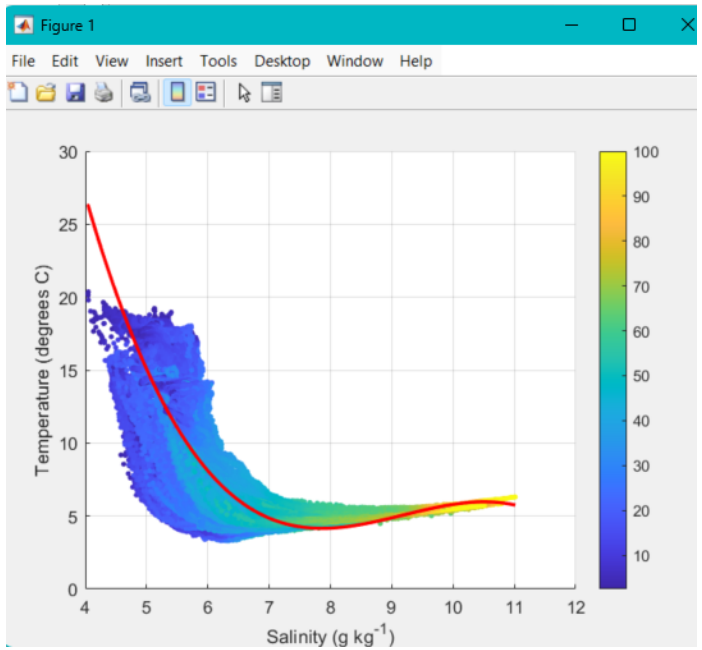
Create temperature.

```
T_pred = predict(stats, S_range);
```

Plot the trend line

```
plot(S_range, T_pred, 'r-', 'LineWidth', 2);
```

Add descriptions to your code and save code.

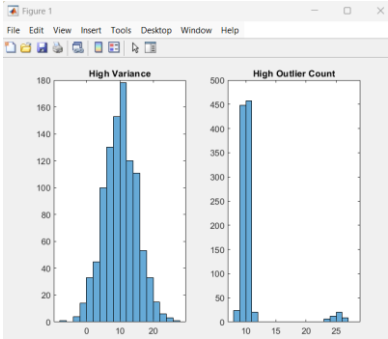


Lecture 5. Exercise 4.

Keri_2020_clean.mat

Let's create a boxchart and visualize salinity data per 10-dbar layers.

- Define bounds for 10-dbar layers.
- Create a variable to hold depth group info.
- Create a variable to hold names of depth groups (e.g., "0-10 m" etc.)



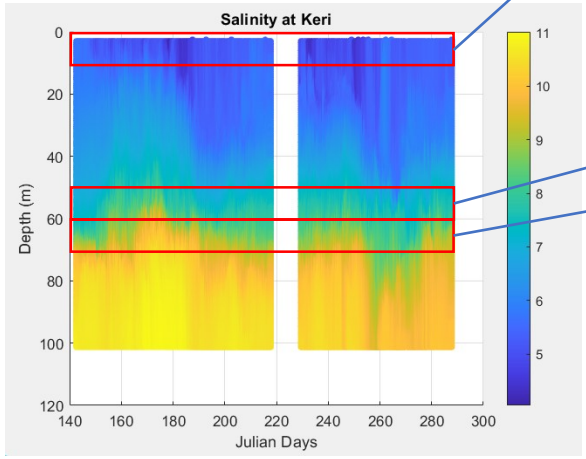
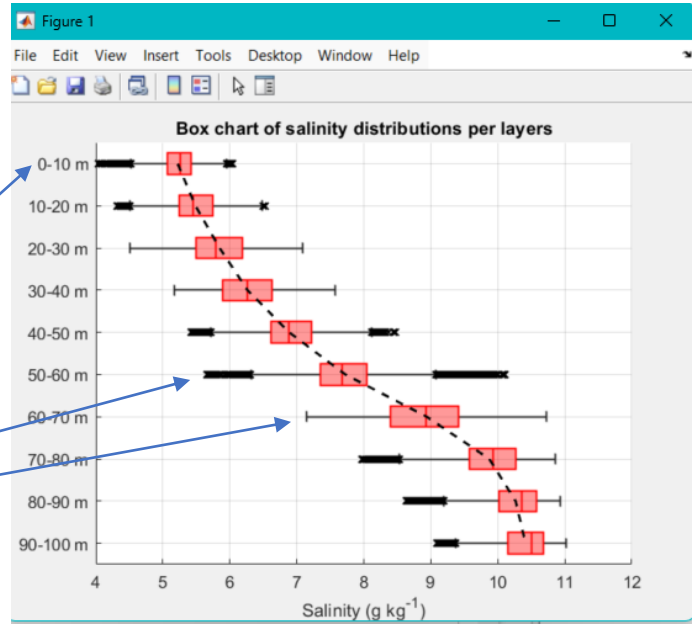
Create boxchart and modify it

- boxchart(dep_group,salinity)
- BoxFaceColor, BoxFaceAlpha, MarkerStyle, MarkerColor, Orientation (horizontal)

- Ydir 'reverse'
- grid on, hold on

- Get mean salinity per depth group and plot it. Use groupsummary

Add descriptions to your code and save code.

How to read a Boxchart:

- The Central Line: The Median value.
- The Box Edges: The 25th and 75th percentiles (the Interquartile Range, or IQR).
- The Whiskers: Extend to the most extreme data points that are not considered outliers (usually 1.5 x IQR).
- Individual Points: Outliers (data that is mathematically far from the rest of the group).